

Perspectives on Urban Walking: Exploring Perceived and Objective Accessibility in Mode Selection

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Motivation

- Planning applications often use objective measures of accessibility to associate policies with specific outcomes.
 - However, objective accessibility may be inconsistent with perceived accessibility.
 - As a result, outcomes may differ from expectations; destinations might be accessible for active travel, but people may not choose those modes.
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Within mid size Spanish urban areas:

- **How does proximity perception and objective accessibility are related to each other?**
- **What is the role of proximity perception in mode choice?**

Case studies: Mid size Spanish Cities

- Granada
- Palma
- Valencia



Data



1. Telephone survey

- ~1200 people (400 in each city)
- 200 in areas of lower accessibility and 200 in areas of higher accessibility

2. Variables

- **Socioeconomics:** Education, age, gender
- **Proximity perception** to different destinations (Shopping, Food, Entertainment)
- **Primary travel mode** for different destinations
- **Residence location**

3. High resolution accessibility data

- Based on residence location
- **Number of shopping, entertainment and food destination within 15 minutes walk**



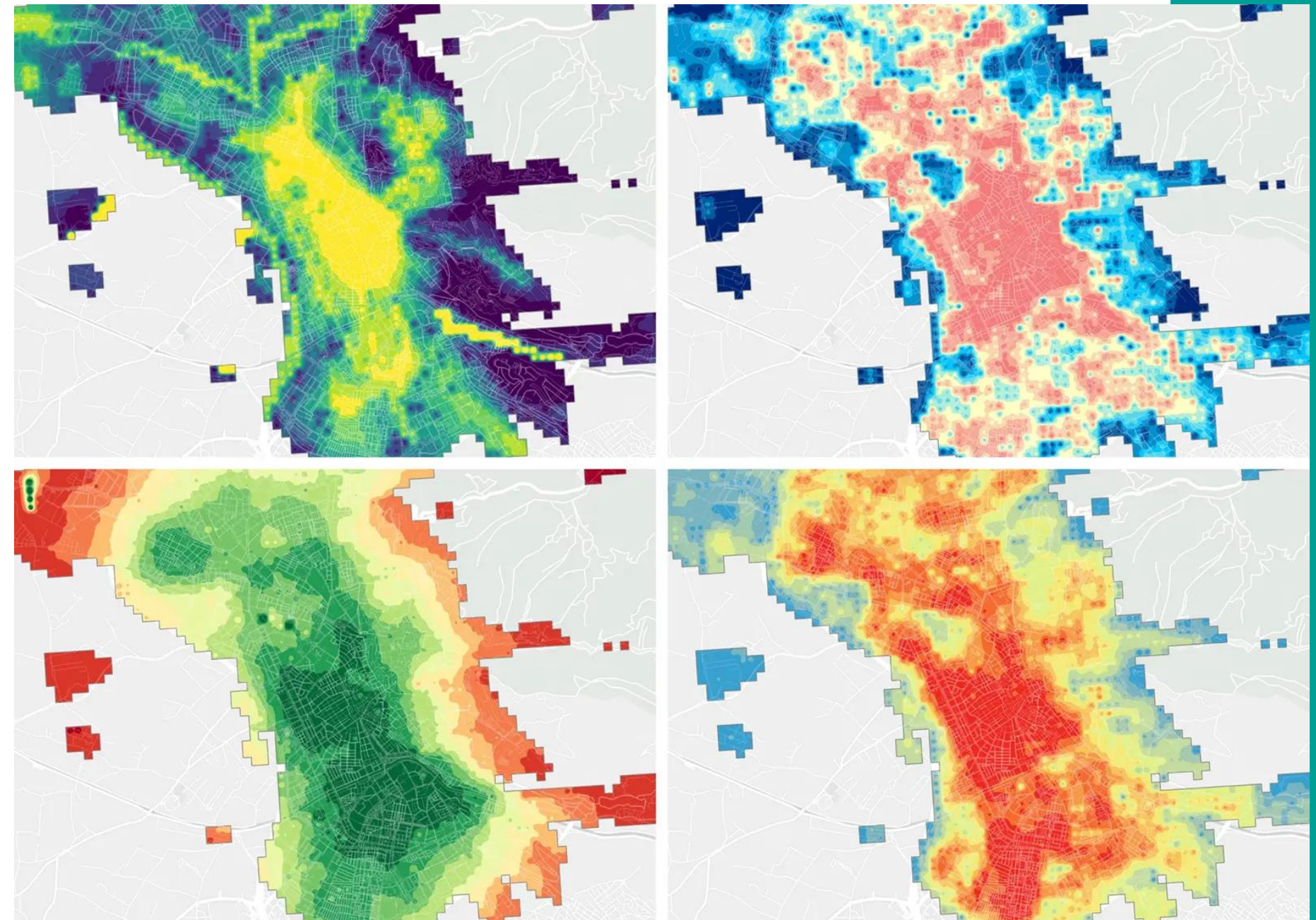
Survey data

Objective Accessibility	Daily shopping	Non-daily shopping	Restaurants	Entertainment
Max	1316	1316	650	226
Min	1	1	0	4
Median	277	279	140	87
Mean	370	370	180	91
Perceived proximity				
Close	94%	78%	85%	71%
Neither Close nor Far	3%	11%	7%	18%
Far	3%	11%	8%	11%

High spatial resolution environmental data

Movactiva data

- Grid of 100mx100m
- 15-min isochrones
- Data from tax records
- Movactiva.es





First Analysis

- Ordered logit: disaggregate choice model that determine the probability of an individual having a certain perception (when they are ordered)
- **Probability of perceived proximity** of certain destinations. Based on the **quantity of destinations** within 15-min walk.
- Four types of destination for all cities
- Analysis consists in seeing the probabilities versus objective accessibility

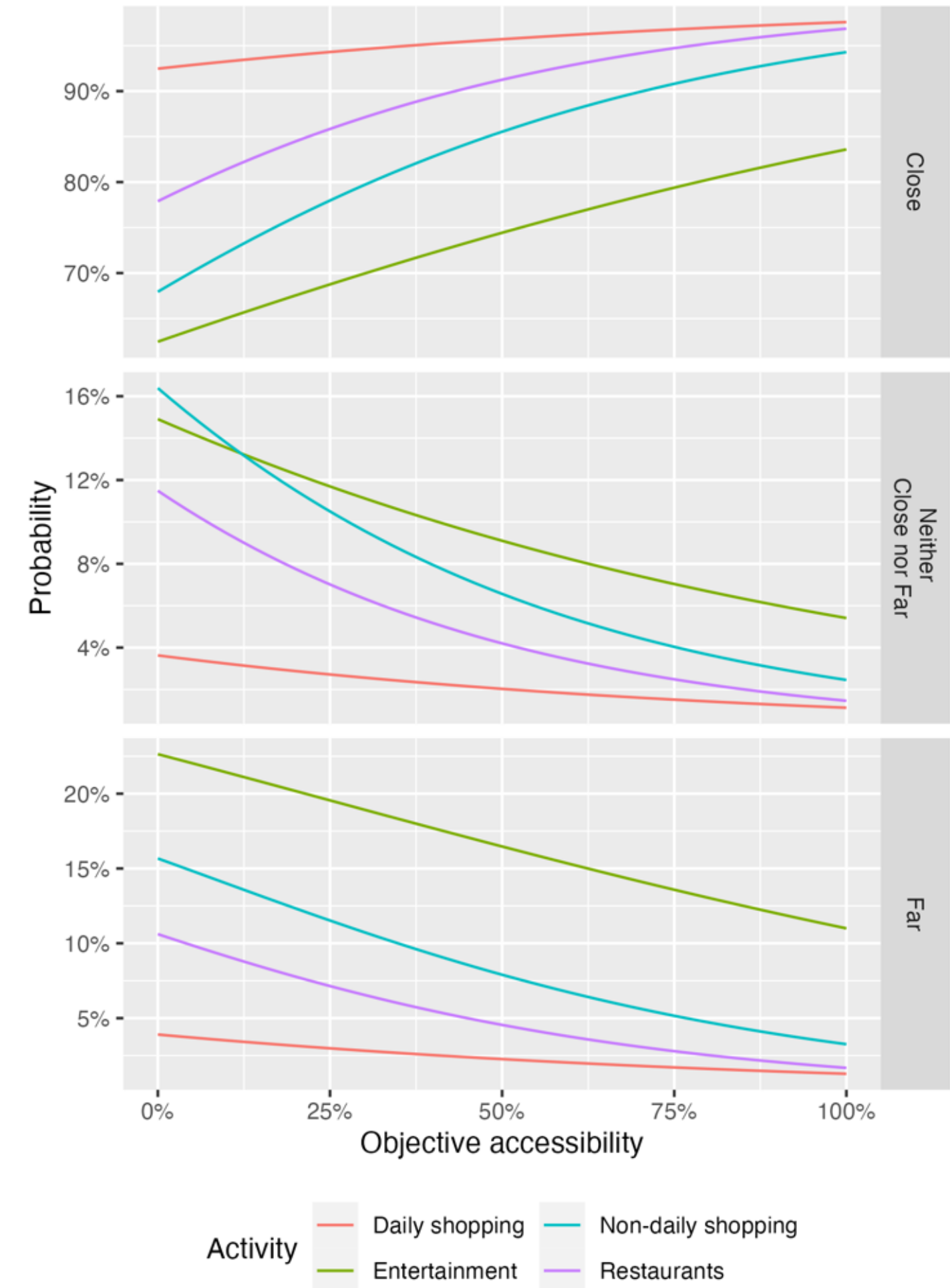
Ordered logit for Perceived Proximity
3-Levels: **Close - neither Close nor Far - Far**
As function of **Objective Accessibility**

1. For **daily shopping**: Shopping destination within 15 minutes
2. For **non-daily shopping**: Shopping destinations within 15 minutes
3. For **restaurants**: Restaurants within 15 minutes
4. For **entertainment**: Entertainment destinations such as theaters or libraries within 15 minutes



Proximity perception probability

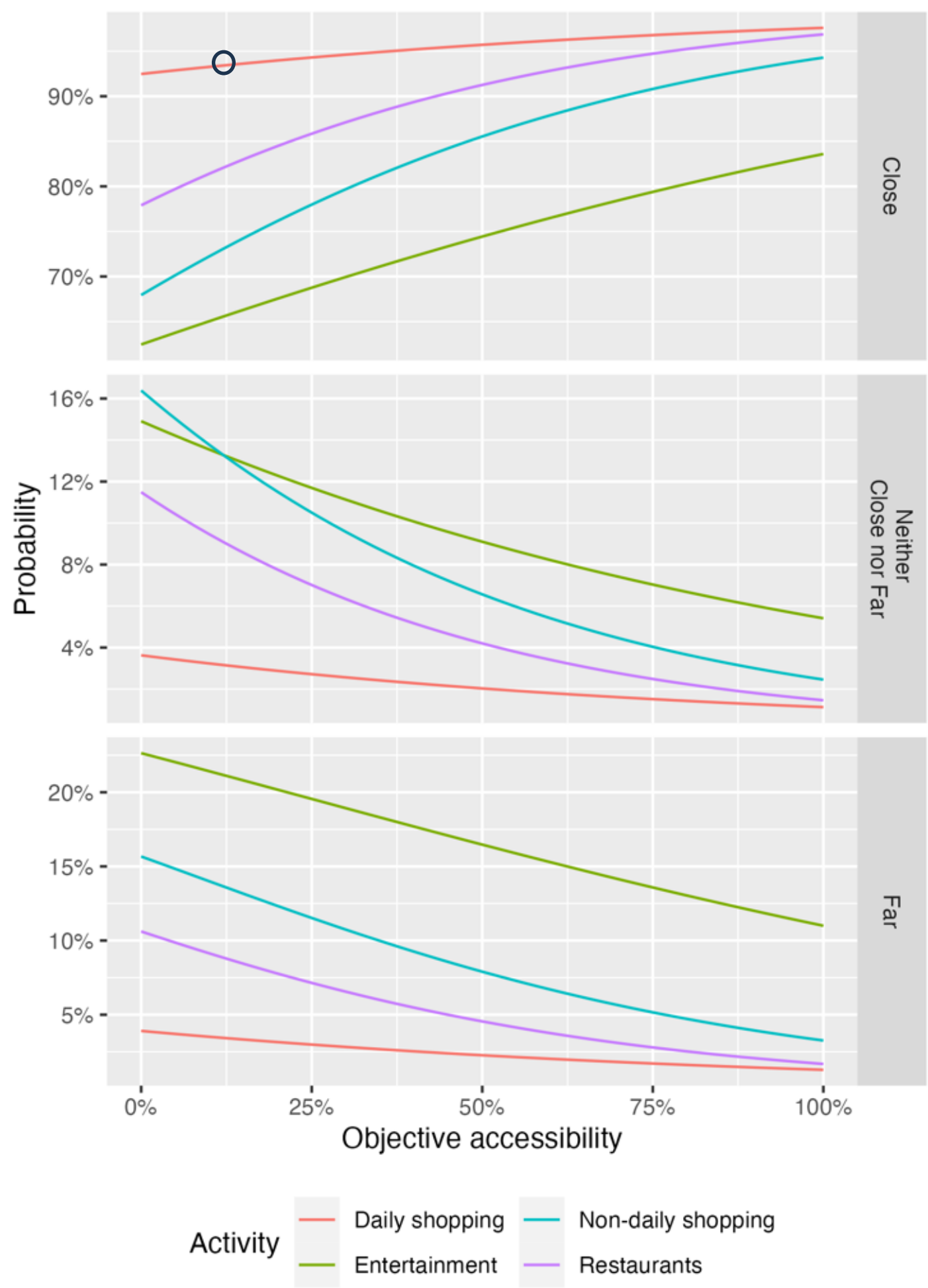
- The perception level is more sensitive to **restaurants** and **non-daily shopping** are more sensitive to accessibility in lower levels.
- These two appear to be more nonlinear than **daily shopping** and **entertainment**.
- In general, in this context, the majority of travelers feel like destinations are close





Proximity perception probability

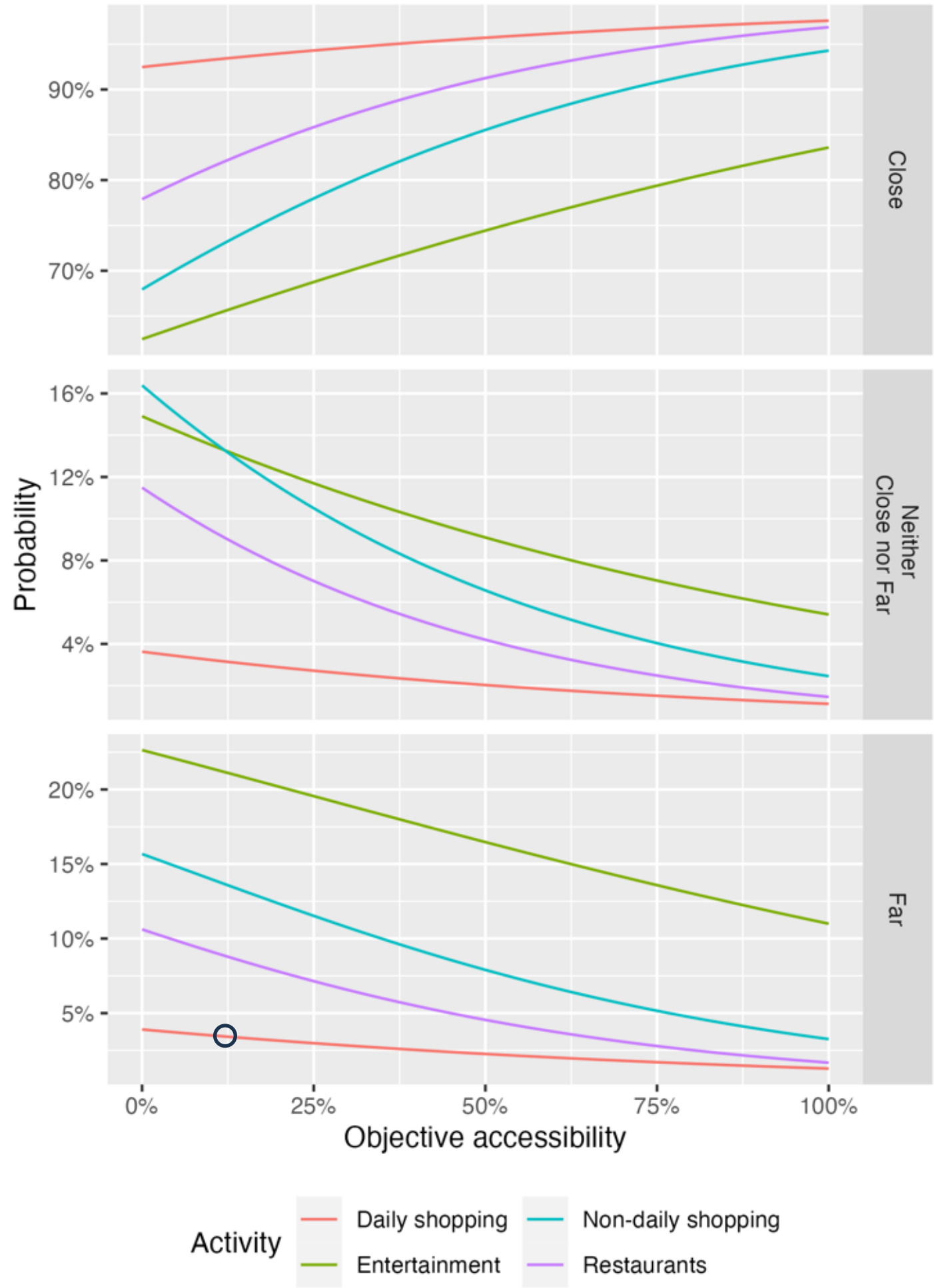
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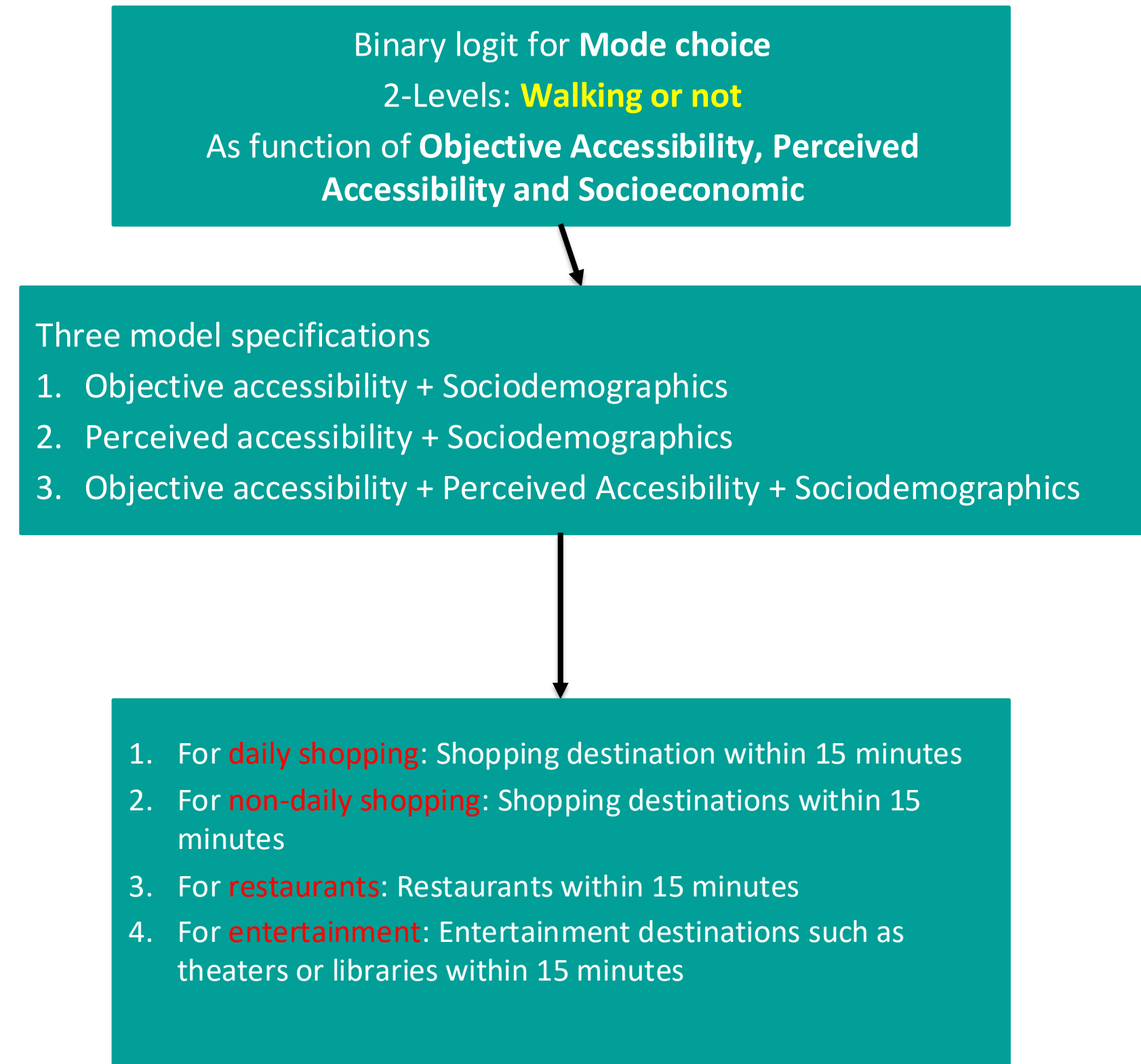
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Second Analysis

- Binary choice model between Walking and other mode
- Three specification
- Four types of destination in all three cities
- Test goodness of fitness across models types within each group of destinations
- Test marginal effects of perceived and objective accessibility across model types

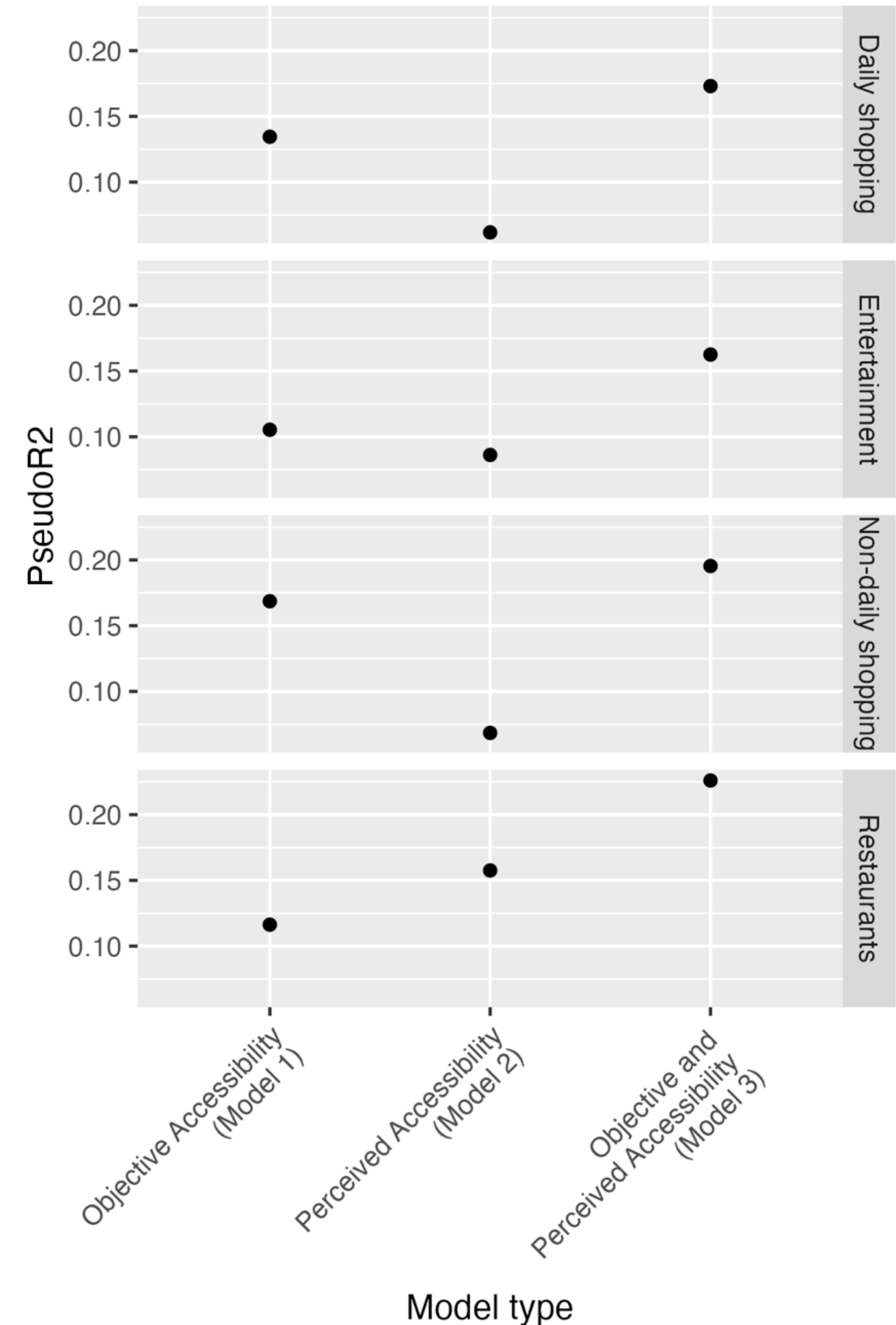




Mode choice

Dependent in perceived and objective accessibility

- Model 3 with both objective and perceived accessibility yield better fit
- Objective accessibility is more explanatory in shopping trips (either non-daily and daily) and to a lesser degree for entertainment
- Perceived accessibility appears to be more relevant than Objective accessibility for the case of cars.

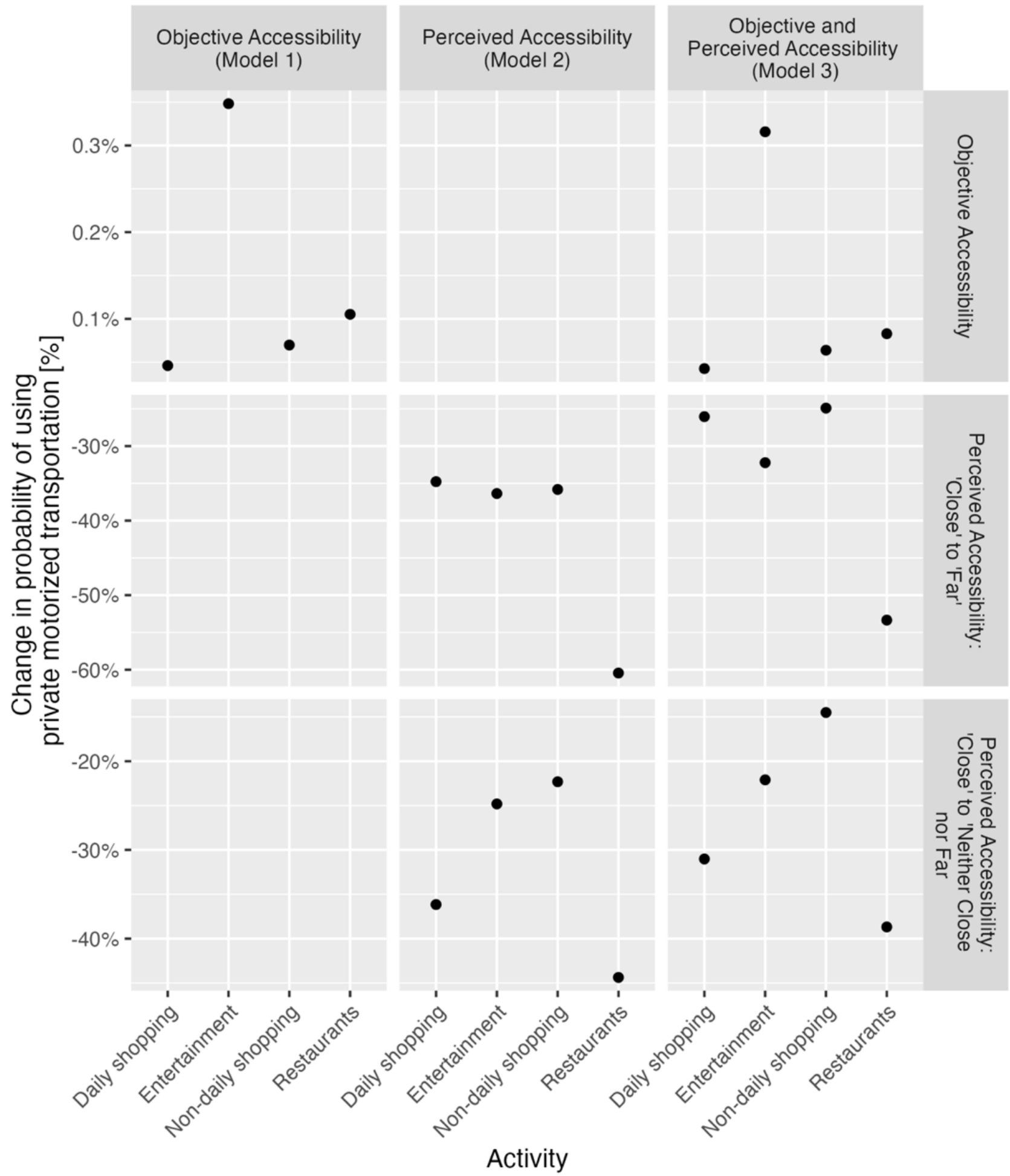




Average Marginal Effects

Change in walking probability due to changes in one unit of the independent attribute

- For example, a change in 1 shopping destination is associated with an increase of 0.05% increase in walking probability.
- When objective accessibility is not included in the model, perceived accessibility becomes more relevant.
- This effect is not as clear when objective accessibility is included but not perceived accessibility.



Discussion

- High objective local accessibility tends to align well with high proximity perception.
 - This finding suggests the existence of a certain threshold in objective accessibility where it aligns with high proximity perception.
 - Additionally, perception plays an important role in modal choice.
 - However, objective accessibility tends to have a relatively consistent effect whether perceived proximity is considered or not in the behavioral model.
 - Finally, the relative importance of perceived proximity will depend on the type of activity, as its relevance can change between activity types.
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Thank you!

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